Moving towards Electronic-Marketing: Computerisation of India's Agriculture

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ABSTRACT

Computerising the administration of agriculture is a major component of India's recent public achievements. Creation of a sub-market (e-NAM) for online transactions within the regulated market is now in process, enabling differentiation of markets in terms of computerisation and evaluation of the technology. A comparison of APMC markets with and without e-NAM for the marketing of a perishable crop, tomato, and a more stable crop, onion, reveals dissimilar performances. Implications for farmers selling in markets with e-NAM seem beneficial, as assessed by higher prices, inter-market equity, farmers' preferences and lower price volatility. The study period is recent but rather stable when movements of prices and arrivals are without secular directions despite volatility and seasonality. Tomato, which is more perishable, seemed to gain in the market through e-NAM, with higher prices and the acquisition of inter-market equity through information dissemination, but post-harvest supply pressures require continued redress.

Keywords: e-NAM, market imperfection, ICT, panel data modelling, information

JEL Classification: Q13, D82, O33, C23, D40

1. Introduction

Agriculture remains a critical sector of the Indian economy in which large sections of the working-class population depend on farming for their livelihood. Not only does it retain its prominence in the nation's food security, but the welfare of the producers in the sector has also increasingly grown in importance in the country's political economy and the development process. While farming was once perceived as a way of life, the sector's competitive strength, its equity across sections and regions, and its potential in the international market can hardly be ignored in the contemporary paradigm. Emerging from a highly fragmented and rather inaccessible and backward state, agriculture needed public support that could not be stopped merely with improved inputs to enhance production. Profitability of farming became an important goal, but communication, information, and updated marketing techniques were constraints. When information technology (IT) began to transform economies at large, it offered potent tools for the government to address these weaknesses.

While benefitting immensely from the green revolution (GR) powered by bio-chemical technology, on the economic side, the resulting production abundance was hurting farmers with low prices. Farm production,

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being highly dependent on weather conditions in a tropical monsoonal climate, exhibits seasonality, which tends to be a feature of price movements. Low prices can force producers to exit the business, leading to the failure of critical product markets.

Volatility of prices is a daunting feature. A spurt in supply, more likely at the harvest time, can crash prices, throwing a market into crisis and farmers into distress. Storage is a limited solution, but for perishable products that need to be consumed and processed fresh, it hardly helps. Marketing of agricultural products is, therefore, a challenging function.

With shifting demand in response to the nutritional consciousness of consumers and the rising public cost of grain storage, the government is encouraging the cultivation of horticultural products, which are expected to generate higher incomes in the farm sector and create secondary income by promoting processing. However, the bulk and proneness to spoilage of agro-products remain challenges in horticultural marketing, thereby encouraging farmers to grow cereals rather than fruits and vegetables. At a larger social level, inability to sell while fresh causes product wastage, garbage disposal issues and environmental degradation with implied over-use of resources at the production stage.

Information transfers across regions may enable enlargement of the market. The integration of the market of a product helps more of its supplies to meet its demand and possibly to fetch the best prices all in good time. Closer and swifter linkages among markets and between potential sellers and buyers, therefore, offer ways of increasing farmers' income, mitigating farmers' harvest-time distress, serving the ecology and, more broadly, of making agriculture competitive, equitable and sustainable.

The government has, in the last three decades, invested heavily in roads in regulated agricultural markets (APMC markets), warehousing and cold chains that eased marketing and helped in drawing private investment in processing. Marketing reforms progressed even with political hurdles inevitable in a federal economy. Alternative channels with better methods have evolved (Ghosh, 2013), but the historical APMC markets remain important for the fair and transparent auctions they have been holding since 1928 and the expertise of risk-taking traders (Ghosh, 2013). The auctions make APMC markets especially influential in price discovery in the country. Intuitively, the advent of IT into agriculture can make this role even more valuable.

Following concerns raised after India’s liberalisation over the degradation of agricultural marketing (Ghosh, 2013), the traditional APMC markets were significantly upgraded with computers for recording data and for networking across markets to enable the sharing of price and sale information. Digitisation proposes a federated farmers’ database structure called IDEA (India Digital Environment of Agriculture), which bridges information gaps between the buyer and the seller and the different APMC markets by computerisation. Part of the IDEA, an e-governance portal called AGMARKNET (Agricultural Marketing Information Network), was launched in March 2000 by the Government of India. Enabling web-based flow of daily arrival and price information from wholesale markets nationwide, it helps traders, researchers, government officials and policymakers make informed decisions, but the main goal is to empower farmers with

timely and accurate information and get the best price.

Equity of market power is a hallmark of the notion of perfect competition. Lack of information, known to be a source of market imperfection (Michler, 2020), leads to inequality among farmers. Formation of monopolies, monopsonies, oligopsonies and even failure of many, if not all, businesses in a sector can be other more extreme outcomes that can jeopardise Indian agriculture. Diminishing interest shown by recent generations in taking up farming as an occupation is well discussed (Michler, 2020). While the APMC laws, as evolved from colonial times, were meant to enforce fairness in marketing, success was elusive.

In reality, traders tended to collude and often remained conformists and even outdated in their practices, even as businesses around the world advanced phenomenally. Corruption added to the problem. Producers had limited access to the larger market, even within the country and remained uninformed of the potential markets. IT can play a major role in retaining and promoting agriculture as a viable and sustainable business (Kaipia et al. 2013, Jedermann et al. 2014, Bohtan et al. 2017). Further, the data generated in real time, as well as historical data, helps in forecasting, planning and reviewing commodity chain activities (Tanksale Patidar et al., 2021). IT began to evolve through the 1980s and 1990s, transforming and integrating the world. IT began to evolve through the 1980s and 1990s, transforming and integrating the world. India's commercial economy embraced IT seriously with the advent of broadband in the early 2000s, but its entry into the farm sector was delayed beyond 2010 and realised only with hand-holding and deliberate support of the government. Public policy began to strengthen the APMC markets with IT. Drawing on public investment, computerisation and the internet not only made APMC markets more informed, but the same intelligence spilling over to other channels created an informed environment. Proposal of a data-stack is under implementation to make technology-enabled easy access to digital and federated agricultural data possible for the benefit of stakeholders and facilitation of various agriculture-related schemes.

In a further innovation from 2016, APMC markets began to be equipped with an online trading platform called e-NAM (electronic National Agriculture Market), which not only shares data among markets but also provides an option for electronic auctioning as opposed to the open and vocal bids practised for price discovery in the rest of the market. Following a successful demonstration in Karnataka, the e-NAM made the data recorded much more factual, real-time and meaningful. Operating as a sub-market in many APMC markets and utilising and taking a share of their space, facilities and even the services of its traders, the e-NAM, by its own demands, is richer in computers.

2. Objective, Data, and Methods

Embodying computerisation at a higher level with accessories, software, networking, maintenance and human resources, e-NAM would add to the enrichment of the larger APMC market that embeds it. Although the intention is to create a national market for any farm product, e-NAM is still nascent; even where installed, the work is still in progress, and has not reached all the APMC markets. At this stage of evolution, although APMC markets are all computerised by now, the incomplete reach of e-NAM could help differentiate the markets by their levels of computerisation and extract the effect of IT. This paper examines whether progress

in IT is showing signs of equalising the APMC markets towards forming a 'one' price ideal for a single product, and if it is helping in mitigating the hardships and uncertainties faced by producers in marketing.

The study, conducted at a time shortly after e-NAM was launched, is based on secondary data collected by the Government of India. It focuses on two dominant horticultural commodities in the Indian diet: (i) tomato in Uttar Pradesh (UP) and (ii) onion in Maharashtra (MH). The choice of the two crops is based on the contrast between the highly perishable nature of the tomato and the relative stability of the onion. Purchased fresh both for direct consumption and processing, the tomato suffers from a short window of harvest and post-harvest period for marketing, while the onion, being relatively more resilient to humidity and heat, lasts longer. Both onion and tomato are multi-season crops. Tomatoes are grown across the country, UP being only one among the growing states. MH, on the other hand, is a major grower of onions, and even within MH, cultivation is concentrated heavily in districts Nasik (12%), Ahmednagar, and Pune.

Samples of APMC markets were selected randomly from across each state, UP and MH, ensuring that the data is available without gaps and about half of the sample markets in each crop-state case have e-NAM. From each market, a time series of monthly data was compiled for the period January 2018 to December 2024, constituting a total of 84 observations. The methodology consists of graphical illustrations and tables to note price movements, marketing preferences, seasonality and inter-market differences in a comparative perspective between two groups of markets: those with e-NAM and those without e-NAM. Taken from the AGMARKNET source, price is measured in value per unit weight (Rs/Quintal) obtained by the seller in the APMC market and 'Arrivals' are expressed in quantity (quintal) units in a market where 100 KG comprises a quintal. Price movement is considered in perspective by comparison with a benchmark of inflation represented by the price index of High-speed Diesel (HSD) from the Office of the Economic Adviser (Website).

Although crop calendars vary within a state, intuitively, both Price and Arrivals can also be subject to seasonality expressed by monthly movements. Decided in open auctions, prices vary from day to day. To avoid bias due to outlying price spikes and missing data (the market is even closed on some days), the modal price is considered the monthly average, but the annual average price is the arithmetic mean of the computed monthly prices. Coefficients of variation (CV) of price are calculated across the sample markets. Prices tend to go down after harvest and shoot up when the crop is in the field. Period of low price (LOW), presumably representing harvest months in the market catchment, and high price (HIGH) are the averages, respectively, of the three lowest and highest annual prices. HIGH and LOW prices are identified using not only numerical assessment but also regression on dummies for months with July as the base, corrected for years.

3. Results

A total of 24 APMC markets are selected for the study of onion in Maharashtra. For tomato in Uttar Pradesh, 20 markets are under study. In both cases, half of the sample markets have e-NAM embedded. Besides rotting easily, a tomato is less dense than the bulky onion. The average price is found to be lower in the markets without e-NAM than in markets with

e-NAM (Table 1). The average Arrivals per market are also more in the markets with e-NAM by over 14 thousand Quintals in the case of onion and by a narrower margin over 1 thousand quintals in the case of tomato. The price advantage could be reflecting efficiency due to access to a larger market enabled by computerisation. Higher arrivals, however, could be reflective of farmers' preferences as well as possibly an intended selectivity of larger markets for setting up e-NAM. Table A1 lists the markets under study, identified as both markets with e-NAM and markets without e-NAM.

Table 1: Sample Market: Number, Average Prices and Average Arrivals in the Sample Period

Markets	Onion			Tomato		
	Number	Price	Arrivals	Number	Price	Arrivals
With e-NAM	12	1715.5	23206.1	10	1704.63	1326.34
Without e-NAM	12	1655.08	8707.19	10	1669.8	185.93
Total	24	1685	15956.7	20	1687.21	761.3

Note: Prices are in ₹/Quintal, and Arrivals are in Quintals/Month in a market. 1 quintal = 100 kg.
*** are prices of onion and tomato per Q so similar? Rs 17/Kg?*****

3.1 Arrivals in Market and Inter-Market Variability

Through the sample period, a mildly increasing trend of the Arrivals in an average market is evident for both crops (Figure 1(a)). For onions, arrivals have grown faster than for tomatoes. Despite volatility, specific seasonality was not marked except that Arrivals tended to rise in the first part of the year. Movements of arrivals are not synchronous between the crops; a sharp peak was noted for onion in March 2023. In July 2023, arrivals of tomato plummeted, even when for onion they peaked. Onion arrivals were more variable than those of tomato, suggesting greater freedom of market disposal, which is not surprising given that onions can be held back longer before sale than tomatoes. Across markets, the CV of arrivals of each crop itself was rather volatile at the beginning of the sample period but stabilised over time, and inter-market price variation was more for tomato than onion (Figure 1(b)).

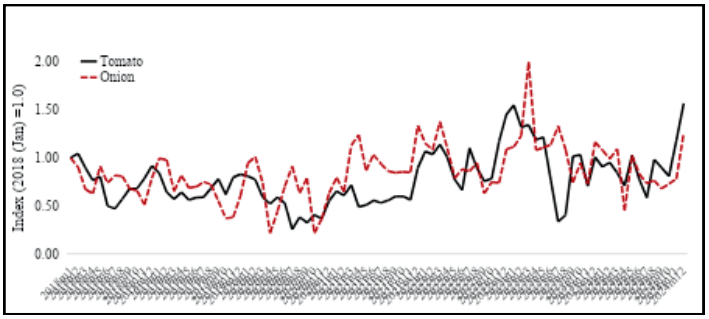


Figure 1 (a): Indices of Arrivals for Markets

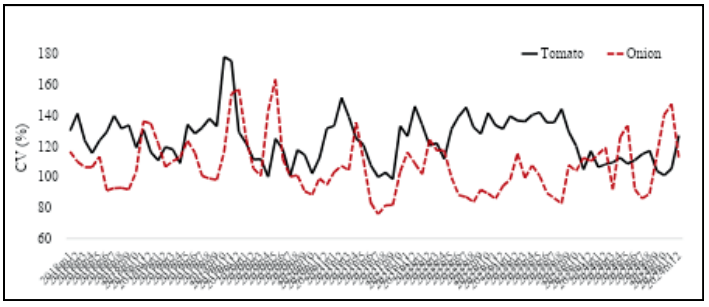


Figure 1 (b): Coefficient of Variation (CV) in Arrivals across the Markets

3.2 Price Movements and Inter-Market Variability

The nominal prices averaged across all sample markets do not show any trend and have not moved with HSD prices that are associated with inflation (Figure 2(a)). Volatility is present in both prices, with a sharp spike observed in the tomato price in July 2023 and in the onion price in October 2019. Tomato prices tend to experience peaks in monsoon months, July and August and onions in the post-monsoon month of November. While any effect of a rise in diesel price is more likely

to hit the consumer than the producer, the price movements relative to January 2018, the start of the sample period, indicate low returns in terms of real prices. Low prices, arguably linked with harvest season, seem to occur in the first few months of the year for both crops.

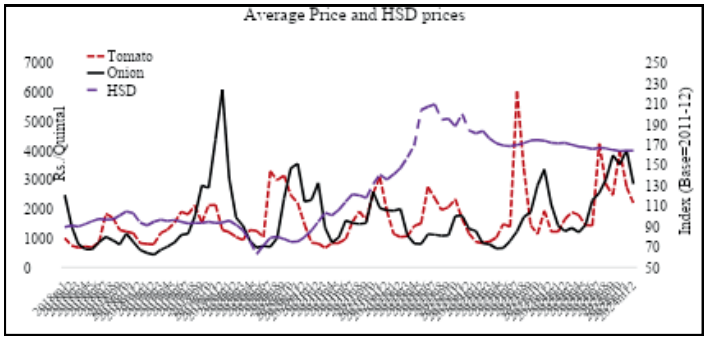


Figure 2 (a): Average monthly market Prices and HSD price index during the sample period

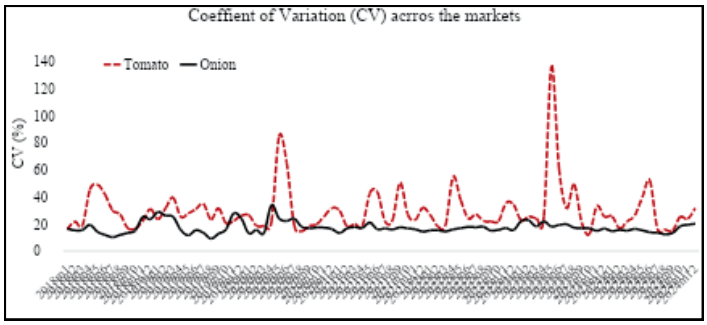


Figure 2 (b): Inter-Market Variation (CV %) of average monthly prices in the sample period

Inter-temporal variability was noted in price movements of both crops (Figure 1(a)), but inter-market price variability holds a different narrative. The graph of CV for onion prices across all sample markets (Figure 2(b)) seems to be decidedly flat compared to that of tomato, even setting aside the outlying spikes. For onions, prices registered are largely uniform across markets and any heterogeneity, if present, has been wiped out over time. Tomato prices have not shown this tendency.

While theoretically perfect competition is associated with uniform prices that make the sellers price takers, in actual practice, the notion rarely meets reality, but performances of food markets are perceived as close to the idea as possible. Comparison to the theoretical ideal and assessment of observed deviations are said to be useful for identifying weaknesses in marketing procedures. For India's agricultural markets, it may be important to mark the deviations for the purpose of addressing inequality among farmers in the catchments of different markets. Admittedly, the idealisation of perfect competition has been critiqued on grounds of inefficiency allowed by the concept (Harvard**).

3.3 Comparison Between Markets with and without e-NAM

Arrivals of onions in a sample APMC market were declining through the period 2018 to 2020, and this was due to a sharp fall in the markets with e-NAM. Onion arrivals in markets with e-NAM picked up pace in 2020-21, even as those in other markets continued growing, but arrivals in both cases slowed down in 2021-22 and again gained momentum in 2022-23. In the last sample year, arrivals in both groups of markets declined, more sharply in markets with e-NAM. Arrivals of tomato behaved likewise. They fell especially in the e-NAM endowed markets up to 2020, picked up pace from 2020 up to 2022 and then declined. In contrast, arrivals in markets without e-NAM largely remained stagnant but grew from 2022, in contrast to markets with e-NAM (Figure 3 (a) and (b)). While arrivals behaved in different ways between the two sets of markets, the e-NAM endowment was more influential on average movements of arrivals.

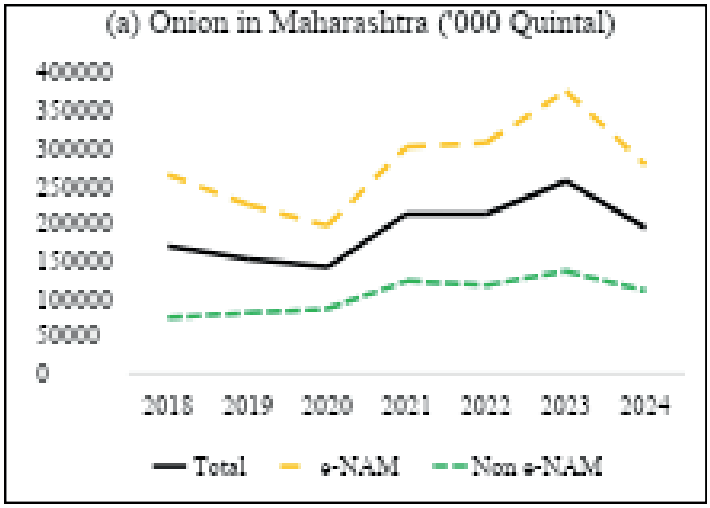


Figure 3 (a): Average Per Market Arrivals for Onion in Maharashtra (Quintal)

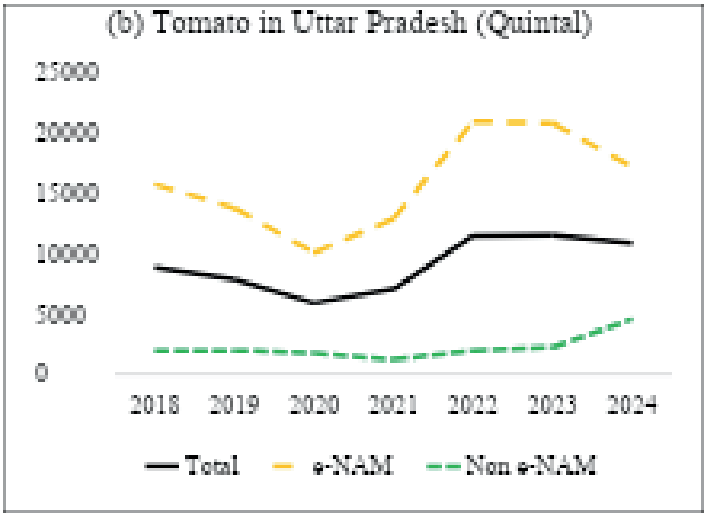


Figure 3 (b): Average Per Market Arrivals for Tomato in Uttar Pradesh(Quintal)

Unlike arrivals, APMC markets differing in their levels of computerisation show largely similar price movements in both cases, indicating information dissemination without bias for or against e-NAM. Prices in markets with e-NAM tend to be higher, more visibly for tomato than for onion (Figures 4 (a) and (b)). While computerisation could be a factor, the differences or lack of them could be accounted for by other reasonable factors. Both markets show volatility and seasonality of prices, and the spikes are sharper in tomato prices.

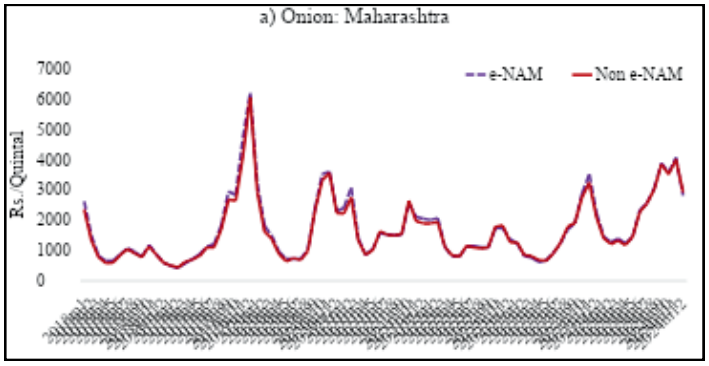


Figure 4(a): Crop-wise onion prices in markets with e-NAM and without e-NAM

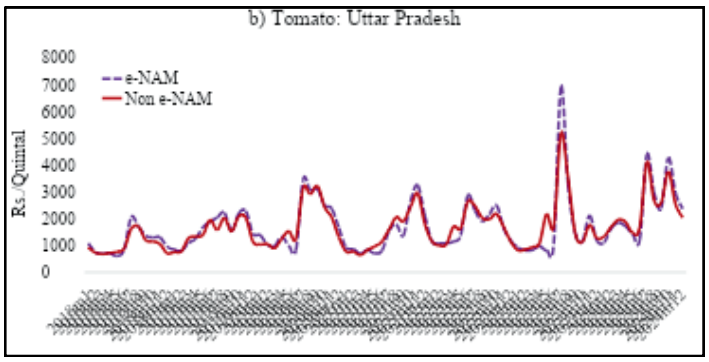
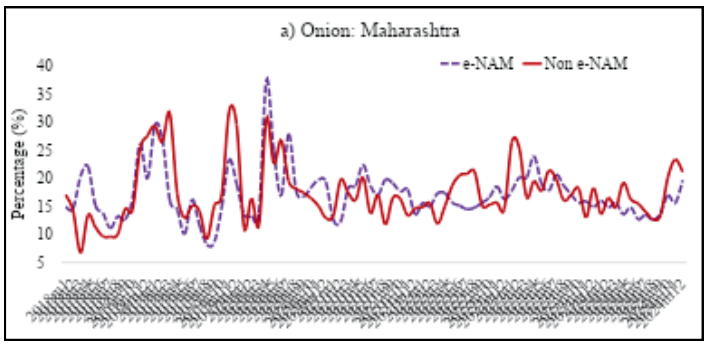


Figure 4(b): Crop-wise tomato prices in markets with e-NAM and without e-NAM

Uniformity of prices among markets (with or without e-NAM) is consistent with inter-market price variability that was observed in Figure 1(b). Variability of prices among different e-NAM possessing markets is, however, relatively strong compared to the other set of markets for onion, but prices are much more uniform across markets with e-NAM than without e-NAM in the case of tomato (Figure 4(a) and (b)). The inter-market CV of price had been observed to be low and flat for onion, and even to that low level of variation, the markets with e-NAM may have contributed significantly, but for tomato, the large inter-market differentials were more due to markets without e-NAM than markets with e-NAM. Although the price behaviours noted through graphs leave much to be explained, it is clear that the two types of markets vary in prices and their variabilities.



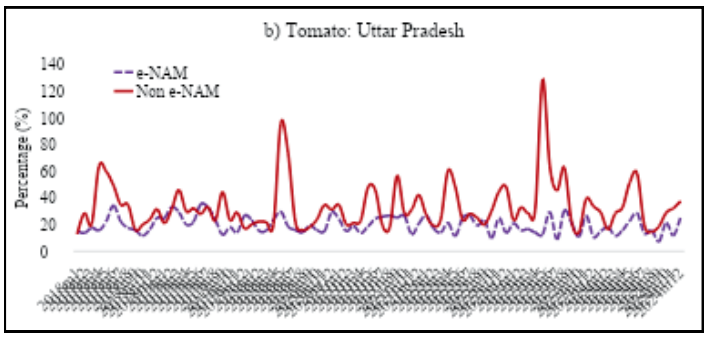


Figure 5(a) and 5(b): Inter-Market Variation (CV %) of Monthly Prices: A Comparison

3.4 Seasonality of Market Arrivals

The analysis so far suggests that the onion prices peak at the end of the year and that of tomato in mid-year. A more rigorous process to identify the patterns would be to regress the prices on dummy variables for months, correcting for years to account for possible trends and larger exogenous shock impacts. Based on an equation estimated from data pooled over 12 months in the sample years and the total number of sample markets using a fixed effect panel data method (Equation A2), three months are identified for the highest prices and three for the lowest prices for each crop. The averages of the prices in each set of three months so identified are classified as HIGH and LOW (Table 2). As with price observations, post-monsoon months are identified in the HIGH category for onion and monsoon months for tomato (Table 2).

It is expected that the LOW price arises due to higher arrivals following a negative price effect of supply and leads to farmers' distress. Implicit is the constraint coming from a rigid demand. On the other hand, the theory of supply suggests a discouraging effect of low prices, which would imply that supplies would increase if prices were higher. Since the production process is already complete, the reality would depend on the ability of farmers to hold on to inventory that might also require better and even technologically refined storage facilities, as well as whether the farmers are enabled. Inverse relations between arrivals representing supplies and prices seen in the scatter diagrams (Figure 6) highlight the importance of demand.

Table 2: Seasonal Pattern of High and Low Prices for Selected Vegetable Crops

Crops	High Price Months (HIGH)	Low Price Months (LOW)
Onion	October, November, December	March, April, May
Tomato	July, August, October	January, February, March

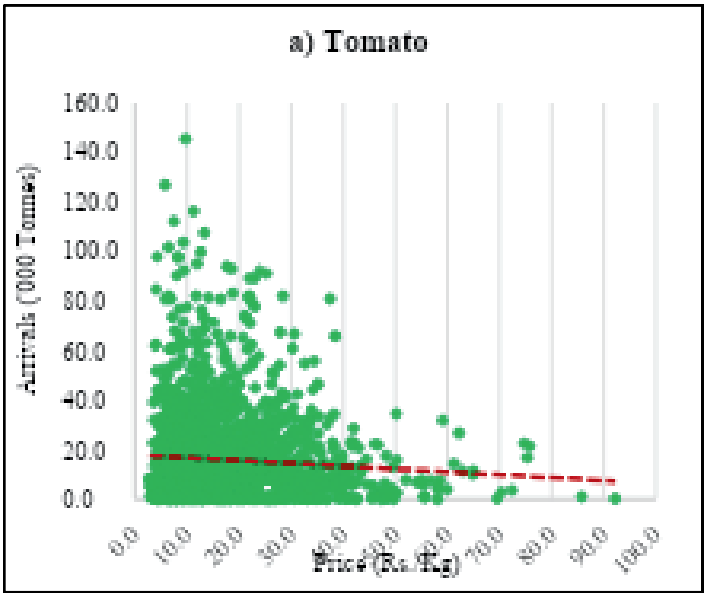


Figure 6(a): Scatter for Tomato Arrivals and Prices

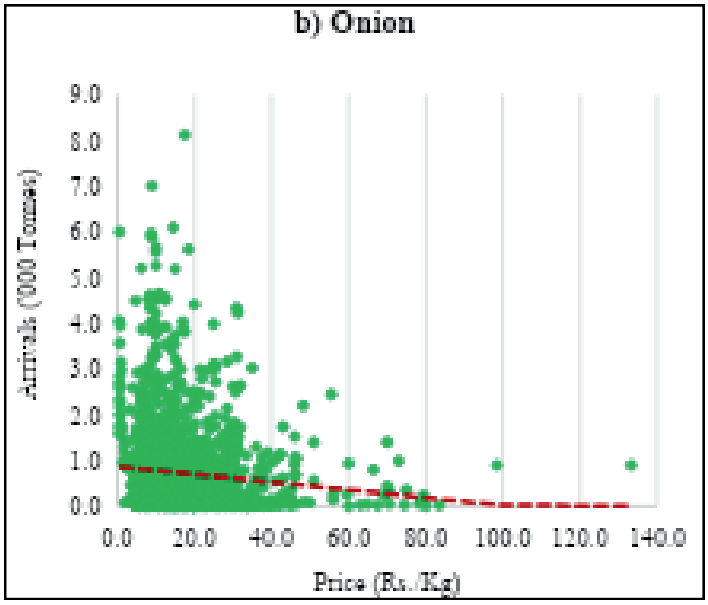


Figure 6(b): Scatter for Onion Arrivals and Prices

Except in the years 2021 and 2024, onion arrivals are higher in the LOW price period than in HIGH price periods, also reflecting demand constraint, causing supply pressure. The difference is about 6% on average, but went up above 50% in two of the sample years. The supply pressure would be higher without e-NAM.

Arrivals of onions are consistently relatively more in the LOW price period in the APMC markets without e-NAM, where the average difference was 70%, differences in most years exceed 70% and even 80% in 2020. In the markets with e-NAM, arrivals were more in the LOW price period than in the HIGH price period to start with, but supply pressure eased so that farmers actually sold more in HIGH price periods subsequently. In the case of tomato, arrivals in the LOW price period were greater than in the HIGH price period in both markets, barring one year each, and the average difference was 40%. Markets with e-NAM did not show any less supply pressure.

Table 3(a): Seasonal Variation in Average Onion Arrivals in Markets with and without e-NAM in Maharashtra

Year	Markets with e-NAM			Markets without e-NAM			All Markets		
Price	LOW	HIGH	% Difference	LOW	HIGH	% Difference	LOW	HIGH	% Difference
2018	20.2	20.2	0.1	7.2	4.1	76.9	13.7	12.1	13.0
2019	18.2	14.1	29.0	8.6	2.8	211.3	13.4	8.4	58.9
2020	11.0	11.6	-5.3	6.3	3.5	82.7	8.7	7.5	15.0
2021	26.0	28.9	-9.9	14.2	8. 9	59.8	8.7	18.9	-54.2
2022	21.1	23.2	-9.0	12.5	8.9	40.5	20.1	17.6	14.0
2023	28.1	26.7	5.2	15.0	8.6	74.5	20.3	13.4	51.4
2024	19.8	27.4	-27.6	12.1	7.1	71.5	16.0	17.2	-7.2
Average	20.6	21.7	-5.0	10.9	6.2	73.6	14.40	13.6	5.80

Note: Arrivals are in '000 Quintals (100 kg)

Table 3(b): Seasonal Variation in Tomato Average Arrivals in Markets with and without e-NAM in Uttar Pradesh

Year	Markets with e-NAM			Markets without e-NAM			All Markets		
Price	LOW	HIGH	% Difference	LOW	HIGH	% Difference	LOW	HIGH	% Difference
2018	1.7	1.0	66.4	0.2	0.1	38.3	0.9	0.6	63.4
2019	1.1	1.15	-2.3	0.2	0.1	44.8	0.7	0.6	2.7
2020	1.2	0.6	107.4	0.2	0.1	139. 7	0.70	0.3	111.8
2021	1.2	1.0	21.7	0.1	0.1	-33.6	0.6	0.5	15.3
2022	1.9	1.6	15.7	0.2	0.1	82.1	1.0	0.9	22.3
2023	2.5	1.2	105.3	0.2	0.1	69.9	1.4	0. 7	101.5
2024	1.4	1.2	16.4	0.3	0.4	-10.3	0.9	0.8	10.1
Average	1.6	1.1	41.3	0.2	0.2	31.7	0. 9	0.6	40.5

Note: Arrivals are in '000 Quintals (100 kg)

4. Conclusion

Increasing development of the electronic market (e-NAM) in the APMC markets is a component of progress in computerisation in agricultural marketing. A comparison of APMC markets with and without e-NAM found similarities as well as dissimilarities between onion and tomato. Both crops are horticultural, but onion is much more resilient to weather, while tomato perishes fast, creating pressure for marketing within the short harvest window. Monthly arrivals in an average market show mild trends with volatility, but inter-market variations of arrivals are greater for tomato.

The average price is found to be lower in the markets without e-NAM than in markets with e-NAM,

indicating that farmers have been able to fetch higher prices with computerisation. There is no bias in price movements for or against e-NAM. While the price advantage probably reflects the efficiency conferred by access to a larger market, other inherent and locational causal factors cannot be ruled out. Arrivals behaved in different ways between the two sets of markets, but larger arrivals in markets with e-NAM are a sign of farmers' rational preferences.

Comparative arrivals are higher in a market with e-NAM, particularly in the case of onions. Arrivals hold a negative relation with prices, suggesting the demand constraint and supply pressure in the marketing season, but for onion, the supply pressure is relatively relaxed in a market without e-NAM, while tomato arrivals rise consistently in periods of low price in markets with and without e-NAM. Both sets of markets show volatility and seasonality of prices, the spikes being sharper and the inter-market variability being considerably more in tomato prices. Onion prices are equally variable, if not more, across different e-NAM possessing markets compared to other markets, while tomato prices are highly uniform across markets with e-NAM but remarkably variable across those without.

Appendix

Table A1: Sample Markets

CROP	Markets	District	Market Type	CROP	Markets	District	Market Type
Onion	Nandgaon	Nashik	With e-NAM	Tomato	Bareilly	Bareilly	With e-NAM
Onion	Rahuri	Ahmednagar	With e-NAM	Tomato	Khalilabad	Sant Kabir Nagar	With e-NAM
Onion	Solapur	Solapur	With e-NAM	Tomato	Jaunpur	Jaunpur	With e-NAM
Onion	Yeola	Nashik	With e-NAM	Tomato	Aligarh	Aligarh	With e-NAM
Onion	Newasa	Ahmednagar	With e-NAM	Tomato	Kanpur	Kanpur	With e-NAM
Onion	Lasalgaon	Nashik	With e-NAM	Tomato	Meerat	Meerat	With e-NAM
Onion	Ahmednagar	Ahmednagar	With e-NAM	Tomato	Gaziabad	Ghaziabad	With e-NAM
Onion	Pune	Pune	With e-NAM	Tomato	Lucknow	Lucknow	With e-NAM
Onion	Chandvad	Nashik	With e-NAM	Tomato	Azamgarh	Azamgarh	With e-NAM
Onion	Satana	Nashik	With e-NAM	Tomato	Gorakhpur	Gorakhpur	With e-NAM
Onion	Vashi	Thane	With e-NAM	Tomato	Muradnagar	Ghaziabad	without e-NAM
Onion	Nagpur	Nagpur	With e-NAM	Tomato	Gazipur	Ghazipur	without e-NAM
Onion	Pimpalgaon	Nashik	without e-NAM	Tomato	Rudauli	Barabanki	without e-NAM
Onion	Manmad	Nashik	without e-NAM	Tomato	Chorichora	Gorakhpur	without e-NAM
Onion	Kalvan	Nashik	without e-NAM	Tomato	Sikandarabad	Bulandshahar	without e-NAM
Onion	Kopargaon	Ahmednagar	without e-NAM	Tomato	Jasra	Prayagraj	without e-NAM
Onion	Niphad	Nashik	without e-NAM	Tomato	Pukharayan	Kanpur Dehat	without e-NAM
Onion	Parner	Ahmednagar	without e-NAM	Tomato	Moth	Jhansi	without e-NAM
Onion	Shevgaon	Ahmednagar	without e-NAM	Tomato	Bijnaur	Bijnor	without e-NAM
Onion	Akole	Ahmednagar	without e-NAM	Tomato	Mehmoodabad	Sitapur	without e-NAM
Onion	Pathardi	Ahmednagar	without e-NAM				
Onion	Amrawati	Amravati	without e-NAM				
Onion	Jamkhed	Ahmednagar	without e-NAM				
Onion	Satara	Satara	without e-NAM				
Tomato	Moth	Jhansi	Non e-NAM				
Tomato	Bijnaur	Bijnor	Non e-NAM				
Tomato	Mehmoodabad	Sitapur	Non e-NAM				

Table A2: Regressions of Price on Month dummies with year effects

Variables	Tomato	Onion
Seasonality	Coeff.	Coeff.
January	-2012.89*** (-21.11)	517.44*** (6.10)
February	-2039.09*** (-21.39)	170.10** (2.00)
March	-1982.89*** (-20.79)	-272.62*** (-3.21)
April	-1832.42*** (-19.22)	-509.93*** (-6.01)
May	-1777.62*** (-18.65)	-463.35*** (-5.46)
June	-1547.07*** (-16.23)	-98.50 (-1.16)
July	*Base*	
August	-608.52*** (-6.38)	240.06*** (2.83)
September	-1083.67*** (-11.37)	711.87*** (8.39)
October	-762.38*** (-7.99)	1,235.91*** (14.56)
November	-887.06*** (-9.31)	1,525.08*** (17.96)
December	-1579.79*** (-16.57)	1,131.71*** (13.33)
Year Trend		
2018	*Base Year*	
2019	483.07*** (6.63)	911.72*** (14.06)
2020	800.13*** (10.98)	803.21*** (12.39)
2021	411.52*** (5.65)	747.74*** (11.53)
2022	646.38*** (8.88)	314.81*** (4.86)
2023	745.54*** (10.24)	507.22*** (7.82)
2024	1269.02*** (17.43)	1,385.21*** (21.36)
Constant	2407.76*** (2.80)	669.21*** (8.19)
Observations	1680	2,016
Number of Markets	20	24

Note: Parenthesis figures are t-statistics. Level of significance at *** is 1%, ** at 5%, and * at 10%.

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